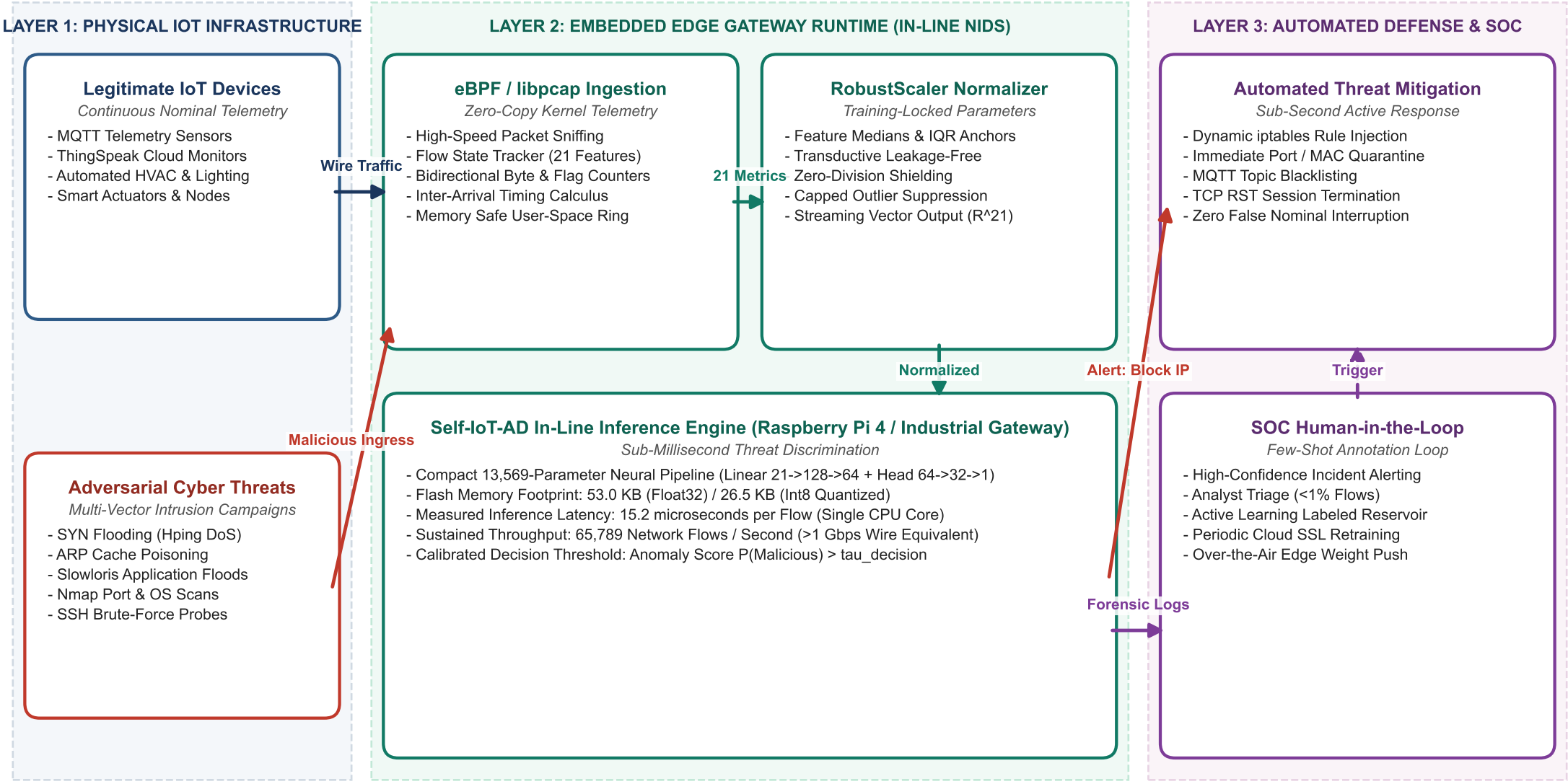

LAYER 1: PHYSICAL IOT INFRASTRUCTURE
LAYER 2: EMBEDDED EDGE GATEWAY RUNTIME (IN-LINE NIDS)
LAYER 3: AUTOMATED DEFENSE & SOC
Legitimate IoT Devices
Continuous Nominal Telemetry
- MQTT Telemetry Sensors
- ThingSpeak Cloud Monitors
- Automated HVAC & Lighting
- Smart Actuators & Nodes
Wire Traffic
eBPF / libpcap Ingestion
Zero-Copy Kernel Telemetry
- High-Speed Packet Sniffing
- Flow State Tracker (21 Features)
- Bidirectional Byte & Flag Counters
- Inter-Arrival Timing Calculus
- Memory Safe User-Space Ring
21 Metrics
RobustScaler Normalizer
Training-Locked Parameters
- Feature Medians & IQR Anchors
- Transductive Leakage-Free
- Zero-Division Shielding
- Capped Outlier Suppression
- Streaming Vector Output (R^21)
Automated Threat Mitigation
Sub-Second Active Response
- Dynamic iptables Rule Injection
- Immediate Port / MAC Quarantine
- MQTT Topic Blacklisting
- TCP RST Session Termination
- Zero False Nominal Interruption
Normalized
Alert: Block IP
Trigger
Malicious Ingress
Self-IoT-AD In-Line Inference Engine (Raspberry Pi 4 / Industrial Gateway)
Sub-Millisecond Threat Discrimination
- Compact 13,569-Parameter Neural Pipeline (Linear 21->128->64 + Head 64->32->1)
- Flash Memory Footprint: 53.0 KB (Float32) / 26.5 KB (Int8 Quantized)
- Measured Inference Latency: 15.2 microseconds per Flow (Single CPU Core)
- Sustained Throughput: 65,789 Network Flows / Second (>1 Gbps Wire Equivalent)
- Calibrated Decision Threshold: Anomaly Score P(Malicious) > tau_decision
Adversarial Cyber Threats
Multi-Vector Intrusion Campaigns
- SYN Flooding (Hping DoS)
- ARP Cache Poisoning
- Slowloris Application Floods
- Nmap Port & OS Scans
- SSH Brute-Force Probes
Forensic Logs
SOC Human-in-the-Loop
Few-Shot Annotation Loop
- High-Confidence Incident Alerting
- Analyst Triage (<1% Flows)
- Active Learning Labeled Reservoir
- Periodic Cloud SSL Retraining
- Over-the-Air Edge Weight Push